Nationwide France Apartment Valuation

Phase 4 Data Engineering, Leakage Controls and Chronological Model Benchmark

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Empirical benchmark edition

Abstract

This report evaluates one region-aware CatBoost automated valuation model on 350,399 untouched 2025 apartment sales represented in nationwide Geo-DVF. Development uses 2021–2023 and model selection uses 2024. The selected model reaches EUR 41,936 MAE, 13.50% median absolute percentage error and $R^2 = 0.8490$. It improves MAE by EUR 10,957 relative to a training-only hierarchical geographic median baseline. All comparable-sale features enforce a minimum 90-day information lag.

Keywords: automated valuation model; Geo-DVF; France; CatBoost; historical comparable sales; chronological validation; regional robustness.

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1 Executive assessment

Phase 4 extends the earlier Paris research into one national apartment model. The central design choice is comparability rather than superficial feature parity: Paris-only transport, school, park, noise and building snapshots are not copied into other regions without a nationally uniform source contract. The national model instead uses DVF property attributes, transaction time, hierarchical territorial identity, global geometry and point-in-time comparable-sale evidence.

EUR 41,936	**13.50%**	**66.97%**	**90.28%**
2025 model MAE	2025 median APE	within ±20%	90% interval coverage

Decision status

- The data, benchmark and reporting protocols are fully specified and tested.
- The frozen 2025 result covers 350,399 transactions; no 2025 outcome influenced model selection.
- Regional MAE is interpreted beside MdAPE, hit rate and test-row count.
- The output is an automated indication, not a certified appraisal.

2 Research objective and contributions

The target is total transaction value in euros for an ordinary apartment sale. The study asks whether one model can learn heterogeneous French markets while preserving local evidence and honest temporal evaluation. Its contributions are:

1. chunk-safe processing of the national annual Geo-DVF archives;
2. explicit support for Corsican and overseas territorial identifiers;
3. spherical H3 and Cartesian location features valid outside Lambert-93;
4. local comparables subject to a 90-day information-availability lag;
5. a training-only hierarchical postcode-to-national median baseline;
6. a 2021–2023/2024/2025 chronological benchmark with paired bootstrap uncertainty;
7. region and sufficiently sized department diagnostics;
8. atomic artifacts, checksums, a reusable production bundle and inference CLI.

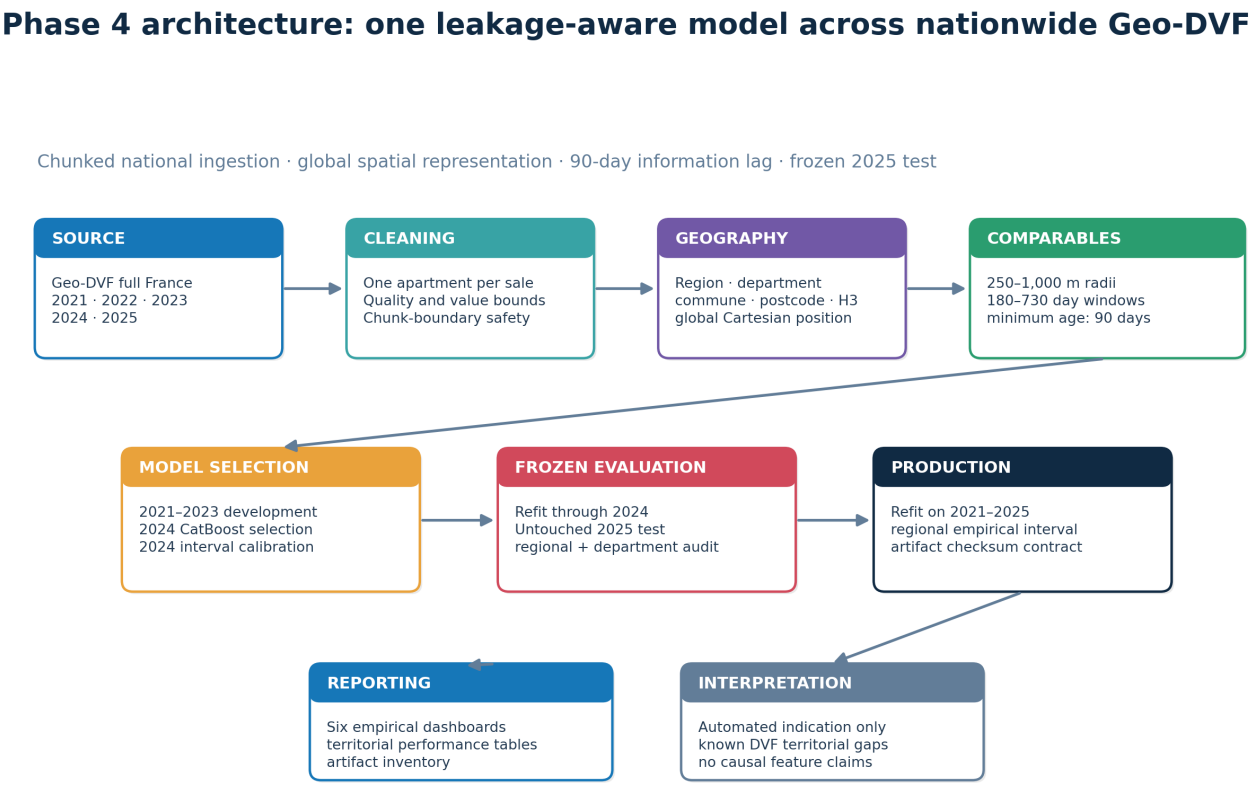


Figure 1: Phase 4 source-to-report architecture and chronological evaluation design.

3 Data source and analytical cohort

The source is the normalized and geolocated Geo-DVF annual full archive for 2021–2025. CSV files are streamed in bounded chunks. The last mutation in each chunk is carried into the next chunk so a multi-line transaction is never classified before all its property rows are available.

The cohort retains *Vente* mutations containing exactly one apartment and no house or commercial premises. Dependencies such as a cellar or parking place may remain bundled in the mutation consideration. Required fields are date, transaction value, apartment surface, commune and valid WGS84 coordinates. Broad national quality bounds are 9–300 m^2, EUR 10,000–10,000,000 and EUR 100–30,000 per m^2.

The audited Gold table contains 1,850,486 sales from 2021-01-01 through 2025-12-31, representing 97 departments and 17 regions.

Geo-DVF contains no open observations for departments 57, 67, 68, 976. “Nationwide” therefore means every territory represented upstream, not every French department without exception.

4 Feature engineering contract

The fitted contract contains 45 inputs: 38 numeric and 7 categorical variables. The current transaction’s target and price per square metre are excluded.

Table 1: Phase 4 feature families.

Family	Inputs	Role
Property	6	Surface, rooms, address number, lots and scale/density transforms
Calendar	4	Seasonality and elapsed market time
Global location	5	WGS84 and unit-sphere Cartesian coordinates
Territorial identity	7	Region, department, commune, postcode and H3 r7/r8/r9
Historical comparables	23	Lagged counts, medians, dispersion, similarity, age, distance and momentum

4.1 Global spatial representation

Lambert-93 is unsuitable for overseas departments. Longitude λ and latitude ϕ are therefore supplemented by unit-sphere coordinates

$$(x, y, z) = (\cos\phi\cos\lambda, \cos\phi\sin\lambda, \sin\phi), \tag{1}$$

and H3 cells. Nearby transactions are queried using haversine distance within each department.

4.2 Leakage-safe comparable sales

For target sale i at date t_i, candidate j is eligible only if

$$90 \le t_i - t_j \le 730 \quad \text{days}, \qquad d(i,j) \le 1{,}000 \text{ m}. \tag{2}$$

Windows at 180, 365 and 730 days and radii at 250, 500 and 1,000 metres produce counts and median prices per square metre. Exponential recency/distance weights, surface and room similarity, effective sample size, dispersion and momentum summarize local evidence. The current transaction cannot become its own input.

Table 2: Primary leakage and integrity controls.

Risk	Enforced control
Mutation target duplication	One row only after transaction-level local-type audit
Future comparable sales	Minimum 90-day age and positive chronological difference
Target-derived leakage	Current value and price/m^2 excluded from fitted inputs
Geographic projection failure	Haversine distances and global unit-sphere coordinates
Model-selection leakage	2024 selection separated from untouched 2025 evaluation
Partial artifact corruption	Sibling `.part` writes followed by atomic promotion
Silent territorial overclaim	Upstream missing departments recorded in every report

5 Benchmark design

5.1 Pre-training audit and market clustering

The empirical preflight covers 1,850,486 retained transactions, 97 represented departments and 17 regions. Weighted lagged-comparable coverage is 94.00%. The median transaction value is EUR 164,000 and median price per square metre is EUR 3,261.

K-means is fitted only to 2021–2023 department profiles. Inputs summarize transaction volume, apartment structure and strictly lagged comparable evidence; the current transaction target is excluded. Median

imputation and standardization apply only inside clustering, not to CatBoost. Silhouette selection retains 2 clusters. These groups are descriptive until the with/without-cluster 2024 ablation demonstrates predictive value.

The best clustered candidate records EUR 42,161 validation MAE versus EUR 42,147 without clustering (difference EUR 14); the cluster feature is therefore not retained.

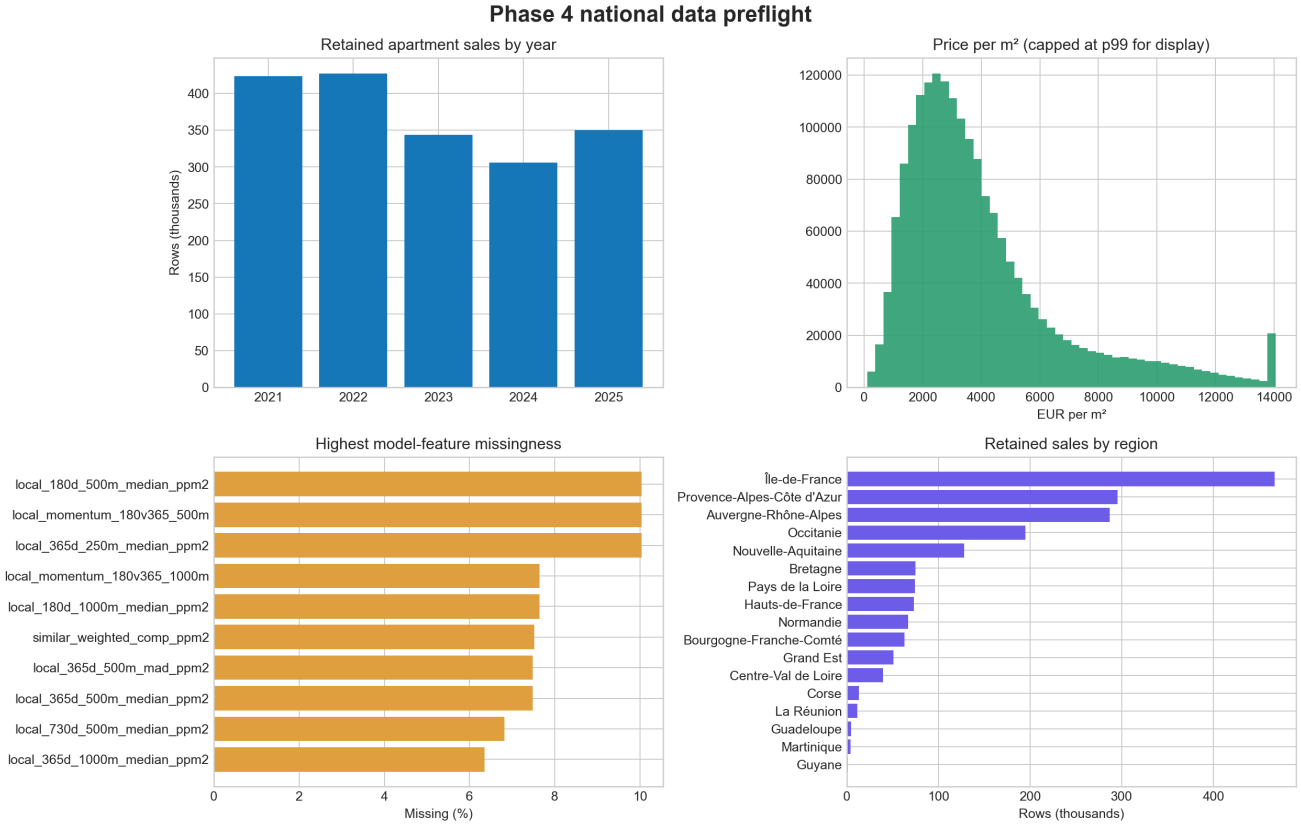


Figure 2: National pre-training audit: annual volume, price distribution, missingness and regional balance.

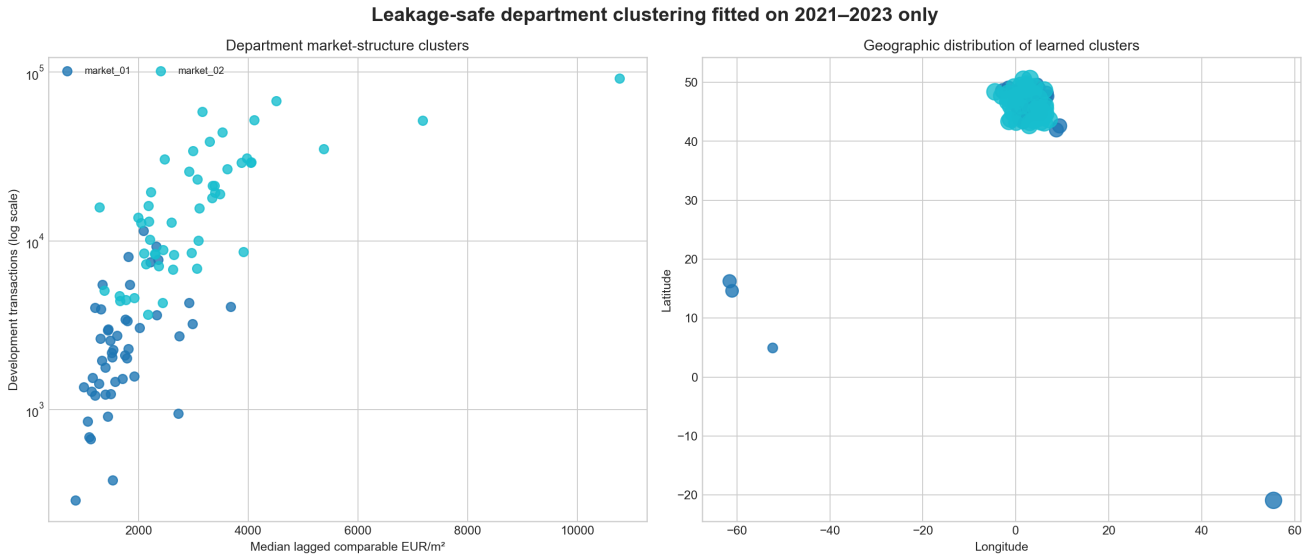


Figure 3: Development-only department market-structure clusters and their geographic distribution.

Development uses 2021–2023. Two predeclared CatBoost configurations are each evaluated with and without the market-cluster feature on 2024; no 2025 outcome participates in selection. The winner is refitted through 2024 and evaluated once on 2025. Only after evaluation is the production model refitted

on all 2021–2025 observations.

Table 3: CatBoost candidate performance on the 2024 selection cohort.

Candidate	Configuration	Cluster	Clipped	MAE	MdAPE	Within ±20%
1	depth=8, loss=MAE, l2=8.0	no	0	EUR 42,147	13.75%	65.55%
3	depth=8, loss=MAE, l2=8.0	yes	0	EUR 42,161	13.74%	65.57%
4	depth=9, loss=Huber:delta=1.0, l2=10.0	yes	111,119	EUR 1,544,040	56.52%	22.83%
2	depth=9, loss=Huber:delta=1.0, l2=10.0	no	201,487	EUR 6,031,220	3289.83%	14.88%

The Huber candidates generated extreme inverse-log outputs. The registered transaction-value bounds prevented non-finite metrics (candidate 2: 201,487, candidate 4: 111,119 clipped predictions), and their resulting validation errors rejected them from selection.

The objective is fitted to $\log(1+y)$ and transformed back to non-negative euros. The benchmark baseline calculates median historical price per square metre by postcode and falls back successively to commune, department, region and national median. Every lookup is learned exclusively from the corresponding training period.

5.2 Metrics and paired uncertainty

The report publishes MAE, RMSE, R^2, MAPE, median absolute percentage error and hit rates within ±10% and ±20%. Because model and baseline predict the same transactions, uncertainty is estimated from paired absolute-error differences. A deterministic bootstrap uses 300 repetitions and at most 250,000 test rows to bound memory. The exact sample size is stored in the benchmark JSON.

6 Frozen national benchmark

Table 4: Frozen 2025 national benchmark on identical apartment transactions.

Method	MAE	RMSE	MdAPE	Within ±20%
Phase 4 CatBoost	EUR 41,936	EUR 92,375	13.50%	66.97%
Hierarchical median baseline	EUR 52,894	EUR 108,830	18.37%	53.65%

The paired MAE reduction is EUR 10,957; its deterministic paired-bootstrap 95% interval is EUR 10,763–11,114.

7 Territorial robustness

Raw euro MAE is expected to be higher in expensive markets. Regional assessment therefore requires MdAPE, hit rates, R^2 and sample size beside MAE. Department rows require at least 100 test transactions; small segments are suppressed rather than presented as stable estimates.

Table 5: Frozen-test performance by region.

Region	Rows	MAE	MdAPE	Within ±20%	R^2
Guyane	315	EUR 19,966	9.22%	80.63%	0.4994
Centre-Val de Loire	7,583	EUR 21,840	13.68%	66.58%	0.7694
Grand Est	9,357	EUR 24,482	16.37%	58.56%	0.7401
Bourgogne-Franche-Comté	11,843	EUR 25,396	16.72%	58.13%	0.7188
La Réunion	2,006	EUR 26,674	12.34%	69.49%	0.8030
Normandie	12,495	EUR 27,043	14.22%	64.66%	0.7758
Hauts-de-France	13,583	EUR 27,640	13.38%	67.53%	0.7954
Occitanie	37,205	EUR 27,735	13.91%	65.27%	0.7473
Pays de la Loire	14,165	EUR 29,952	12.79%	69.22%	0.7783
Bretagne	14,159	EUR 30,070	12.64%	70.00%	0.7033
Martinique	727	EUR 30,893	14.17%	64.24%	0.6421
Guadeloupe	801	EUR 32,700	14.75%	62.92%	0.7218
Nouvelle-Aquitaine	24,590	EUR 33,378	12.85%	68.41%	0.7878
Auvergne-Rhône-Alpes	53,825	EUR 37,479	14.36%	64.55%	0.8171
Corse	2,596	EUR 47,649	16.12%	59.05%	0.6277
Provence-Alpes-Côte d'Azur	56,352	EUR 49,751	15.34%	61.36%	0.7395
Île-de-France	88,797	EUR 62,277	11.71%	73.98%	0.8533

8 Calibration and uncertainty

The empirical 90% interval uses 2024 target-to-prediction ratios. A region gets its own calibration ratios only with at least 500 observations; otherwise it inherits the national interval. Observed 2025 coverage evaluates temporal transfer but is not a formal conditional guarantee for a specific apartment.

9 Visual diagnostic suite

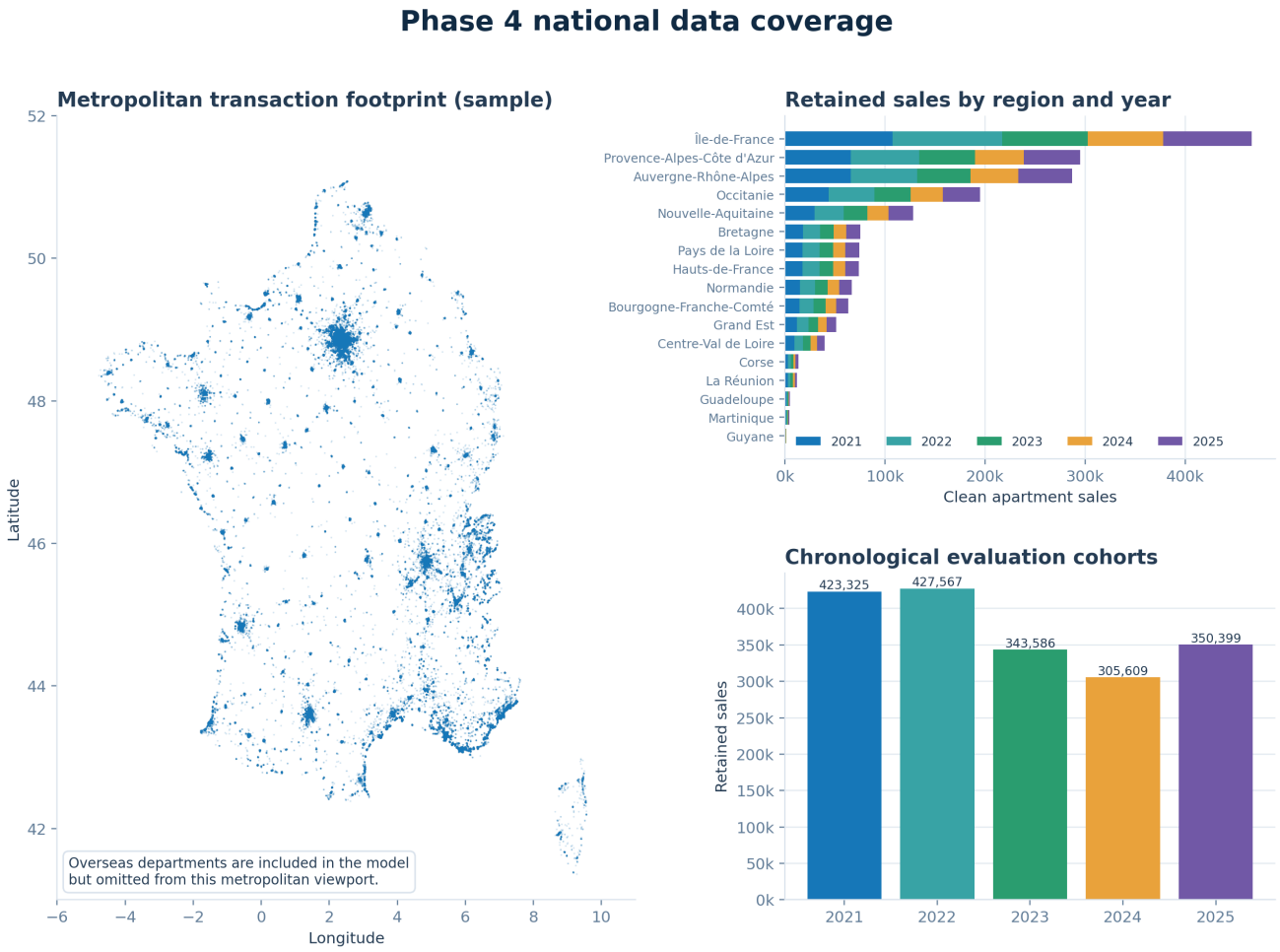


Figure 4: National transaction footprint, retained sales by region and chronological cohorts.

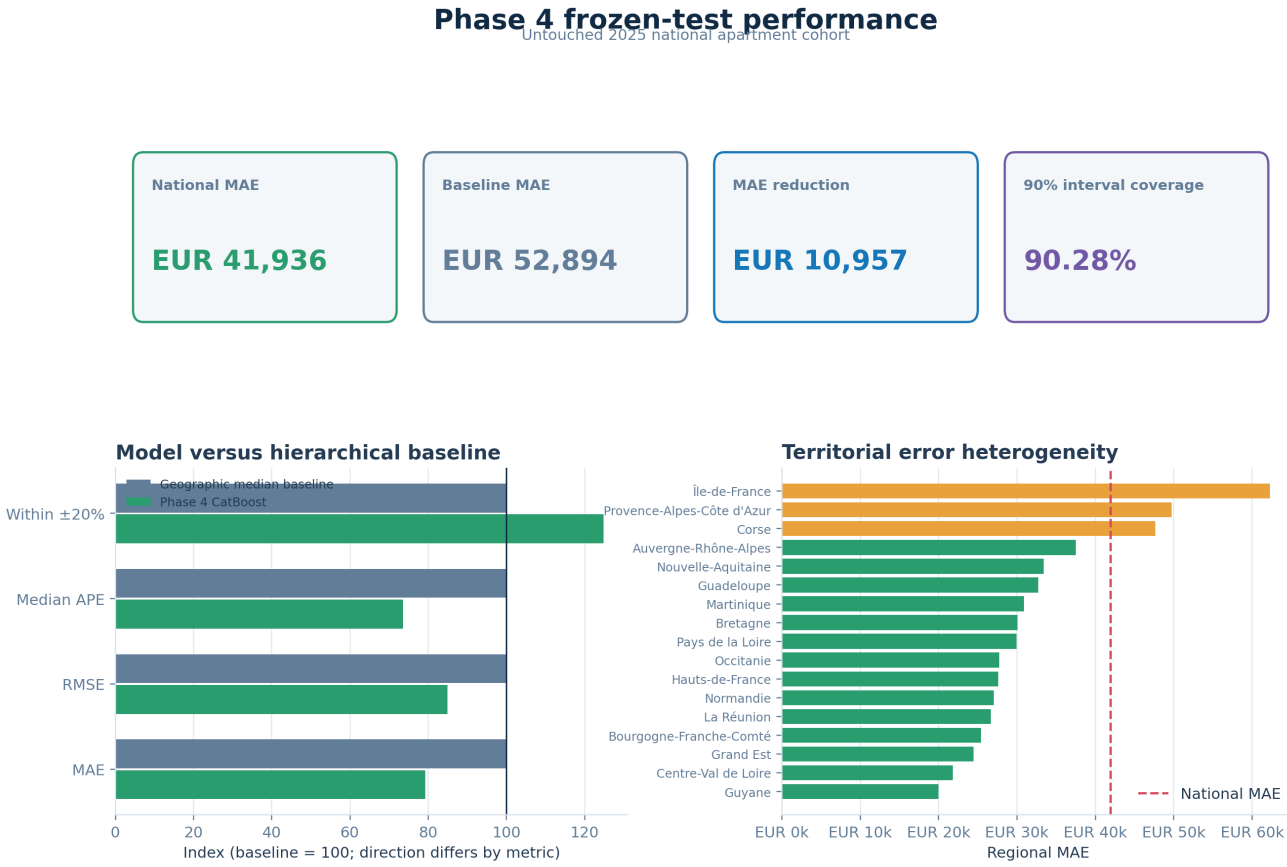


Figure 5: Frozen-test model-versus-baseline benchmark and regional error heterogeneity.

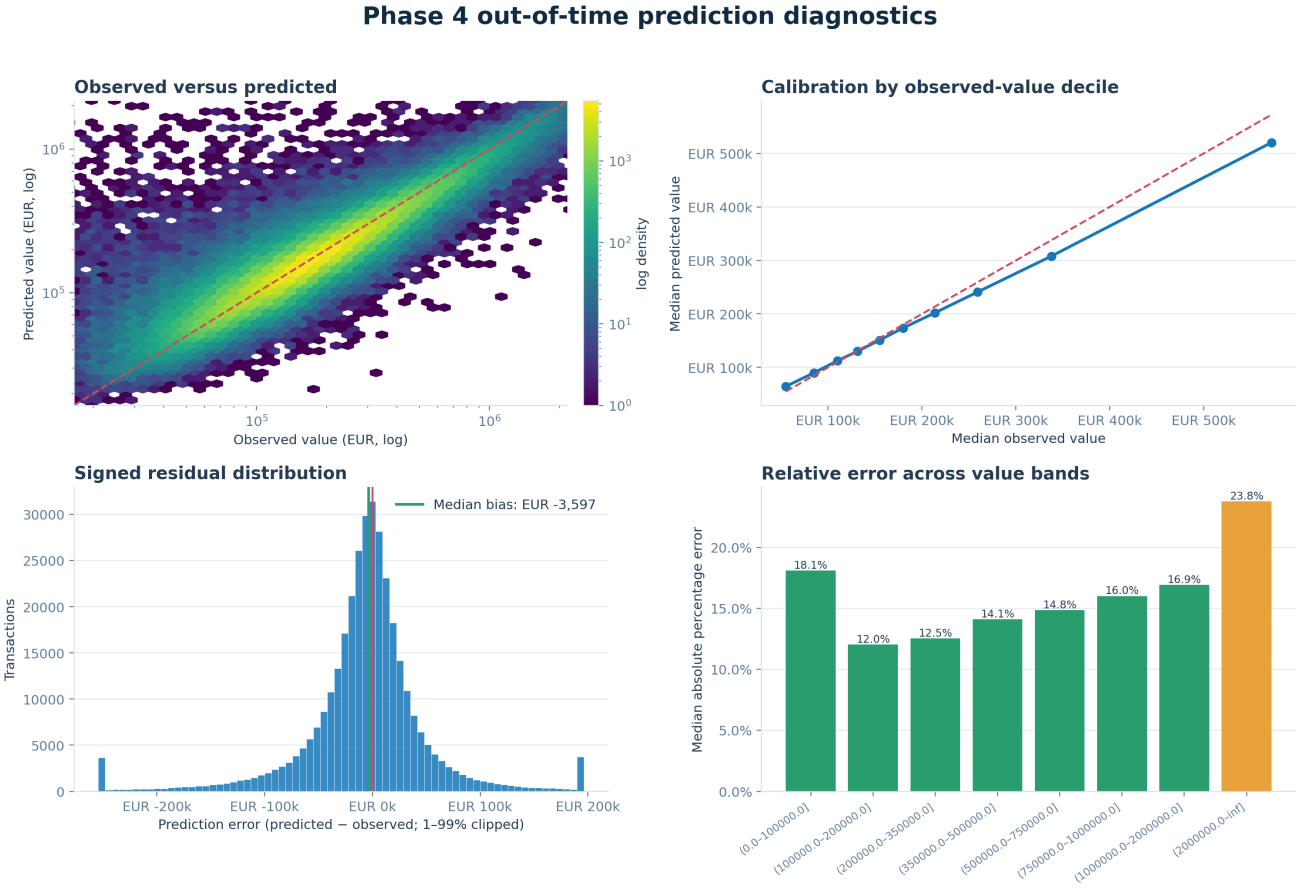


Figure 6: Out-of-time calibration, residual and value-band diagnostics.

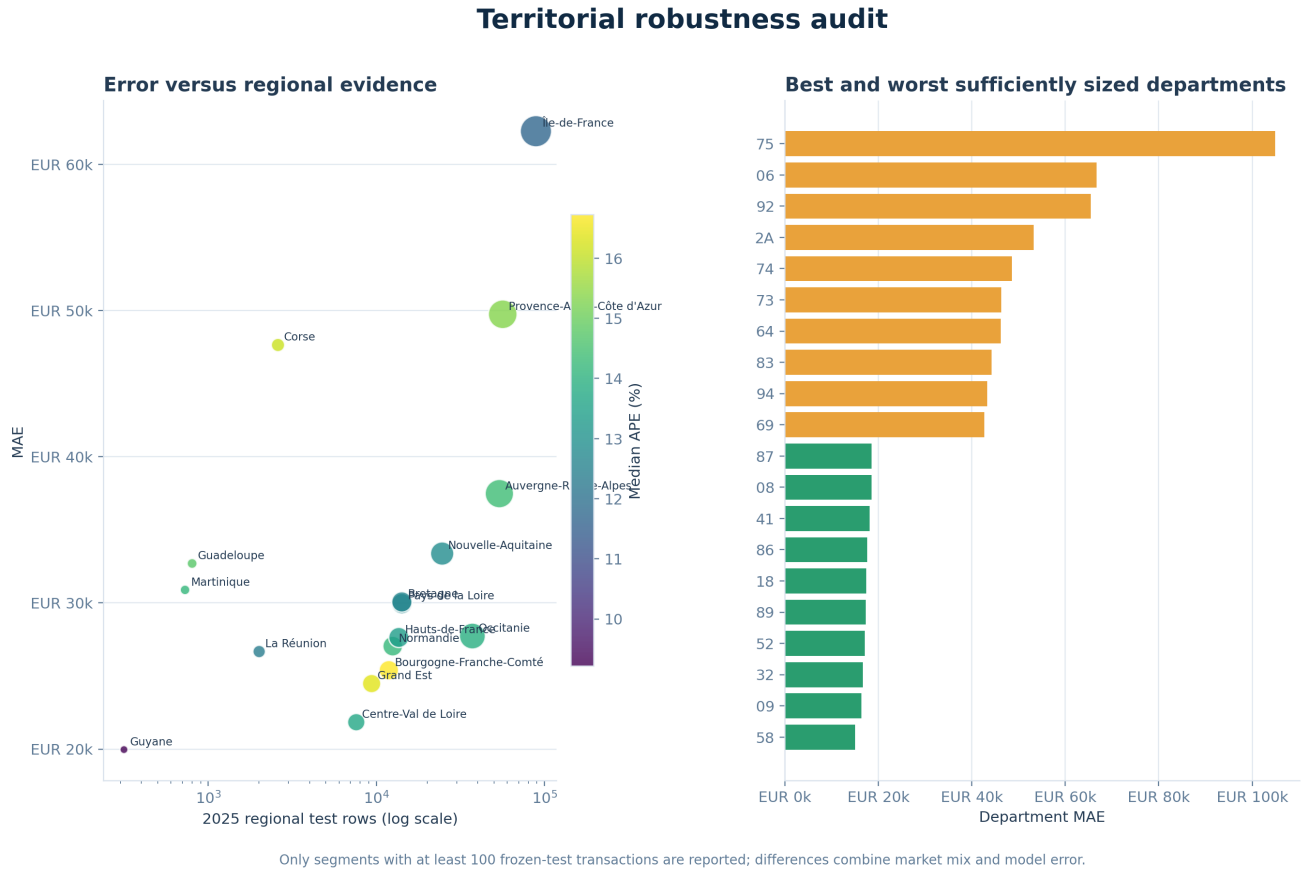


Figure 7: Regional evidence and department-level robustness audit.

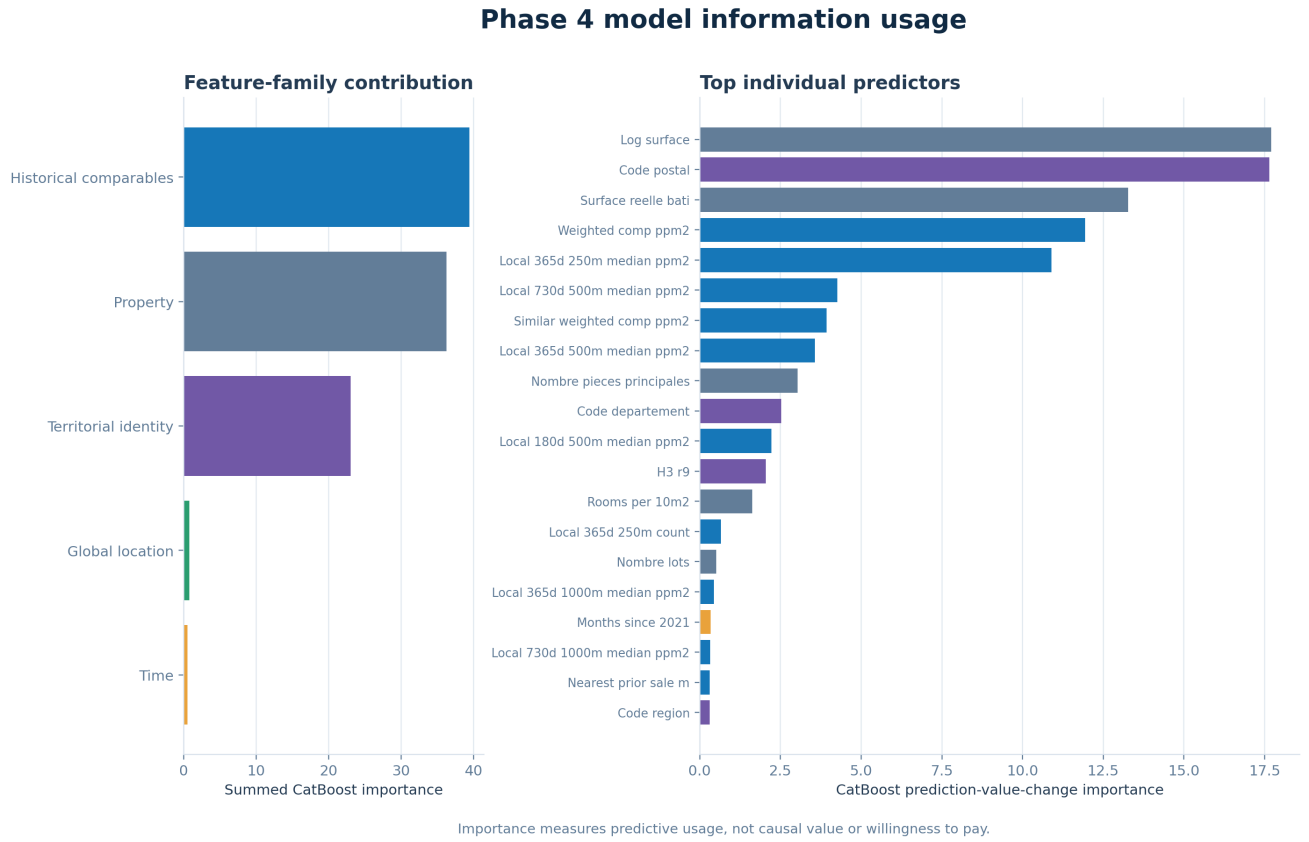


Figure 8: CatBoost information usage by feature family and individual input.

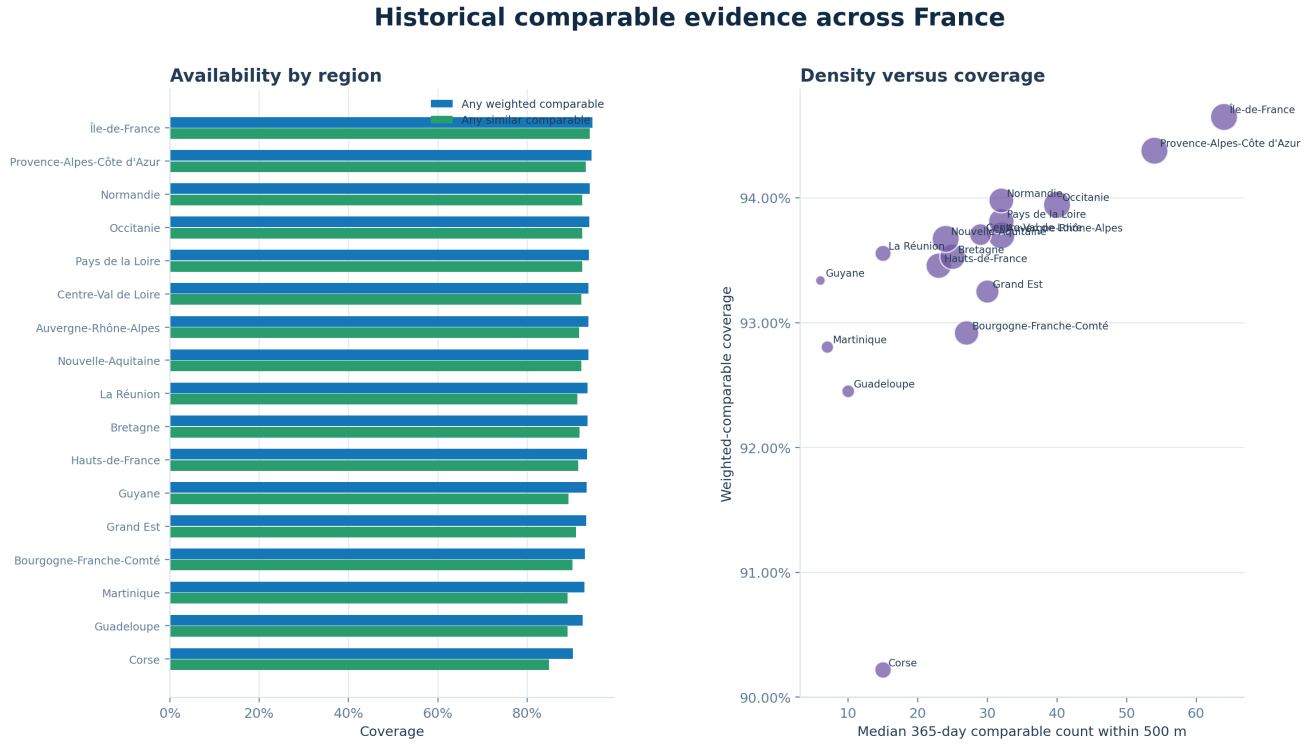


Figure 9: Lagged comparable-sale availability and density by region.

Figures are generated only from existing artifacts. Missing benchmark outputs produce no placeholder chart and no synthetic metric. Feature importance records predictive usage rather than a causal housing premium.

10 Reproducibility and artifact inventory

Artifact	Status	Bytes
Source manifest	available	2,319
Gold feature table	available	340,479,232
Feature quality report	available	8,645
Preflight summary	available	2,931
Department market clusters	available	21,305
Preflight Markdown report	available	2,940
Benchmark JSON	available	7,425
Validation trials	available	817
Benchmark comparison	available	462
Frozen predictions	available	15,829,165
Regional metrics	available	1,399
Department metrics	available	5,795
Feature importance	available	1,778
Production model	available	497,458,679

```
python -m paris_avm.data.acquire_phase4
python -m paris_avm.features.phase4
python -m paris_avm.analysis.phase4_preflight
python -m paris_avm.modeling.benchmark_phase4
```

```
python -m paris_avm.visualization.phase4
python -m paris_avm.reporting.phase4
powershell -ExecutionPolicy Bypass -File .\docs\paper\compile_phase4.ps1
```

11 Limitations and responsible use

- DVF omits condition, renovation, floor, elevator, view and occupancy.
- Mutation value can bundle dependencies or transaction-specific rights.
- Departments 57, 67, 68 and 976 are absent from the source.
- Sparse submarkets may have weak comparable coverage and unstable errors.
- Administrative/H3 identity is predictive context, not a causal estimate.
- A 2025 holdout does not establish performance in 2026 or later markets.
- The interval is empirical and marginal, not a certified property guarantee.
- Predictions are not notarial valuations, lending decisions or investment advice.

12 Conclusion

Phase 4 defines a scalable and auditable path from a Paris research model to a national France AVM. Its strongest contribution is the combination of global geography, local point-in-time market evidence, chronological evaluation and territorial reporting. In this empirical benchmark edition, claims remain limited to the artifacts actually present. The completed frozen evaluation supports a national 2025 MAE of EUR 41,936 and a paired improvement of EUR 10,957 over the training-only baseline. The empirical claims in this edition are generated directly from the registered benchmark artifacts; limitations remain especially important for sparse territories and for applications beyond the 2025 evaluation period.

A Machine-readable deliverables

The benchmark publishes JSON metrics, validation trials, a two-method comparison, regional and department CSV tables, Parquet predictions with paired errors, feature importance and a checksummed Joblib production bundle. The Markdown and LaTeX reports consume these outputs without refitting or selecting a model.

B Source references

- Geo-DVF dataset catalogue on data.gouv.fr
- Geo-DVF annual file index
- Open Licence 2.0